

Modern Technology

Render Queue

(updated w2026)

What is the Render Queue?

- The Render Queue is the updated rendering plugin for Unreal. It allows you to render out high quality frames on animation, and includes features like multi-camera output, and multiple file types.
- It replaces the Movie Scene Capture (Legacy) which is the older rendering module.
- Additional information here:
- <https://dev.epicgames.com/documentation/en-us/unreal-engine/rendering-high-quality-frames-with-movie-render-queue-in-unreal-engine>

The Render Queue

- Add the Render Queue Plugin if it isn't in the project already.

The screenshot shows the Unreal Engine interface with the 'Plugins' window open. The 'Render Queue' tab is selected, displaying a list of plugins. The 'Movie Render Queue' plugin is checked, while 'Movie Render Queue Additional Render Passes' and 'Niagara MRQ Support' are unchecked. The left sidebar shows a list of categories with counts, and the top menu bar includes File, Edit, Window, Tools, and Help.

Category	Count
BUILT-IN	740
2D	1
Accessibility	3
Advertising	1
AI	9
Analytics	4
Android	3
Android Background Service	1
Animation	71
Audio	29

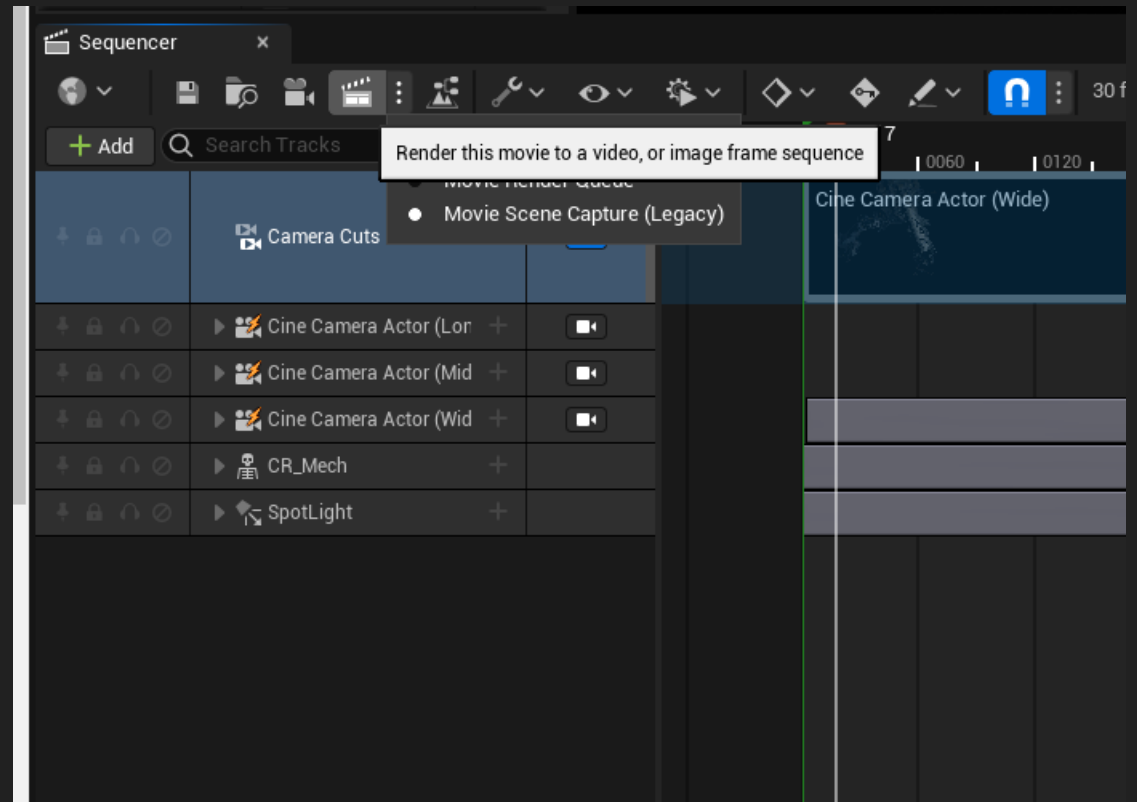
Plugin Name	Status	Description
Movie Render Queue	Checked	Advanced movie rendering pipeline for use in creating rendered cinematics or other multi-media creation.
Movie Render Queue Additional Render Passes	Unchecked	Additional render passes for the Movie Render Queue. This currently includes the ObjectId pass (Editor Only) which generates object mattes with some limitations (using the Cryptomatte specification), and a Panoramic pass with better Sequencer integration than the Panoramic Capture plugin. Beta
Niagara MRQ Support	Unchecked	Contains a data interface that can be used to read Movie Render Queue information in Niagara simulations. Beta

Rendering Video (Old way)

Movie Scene Capture (Legacy)

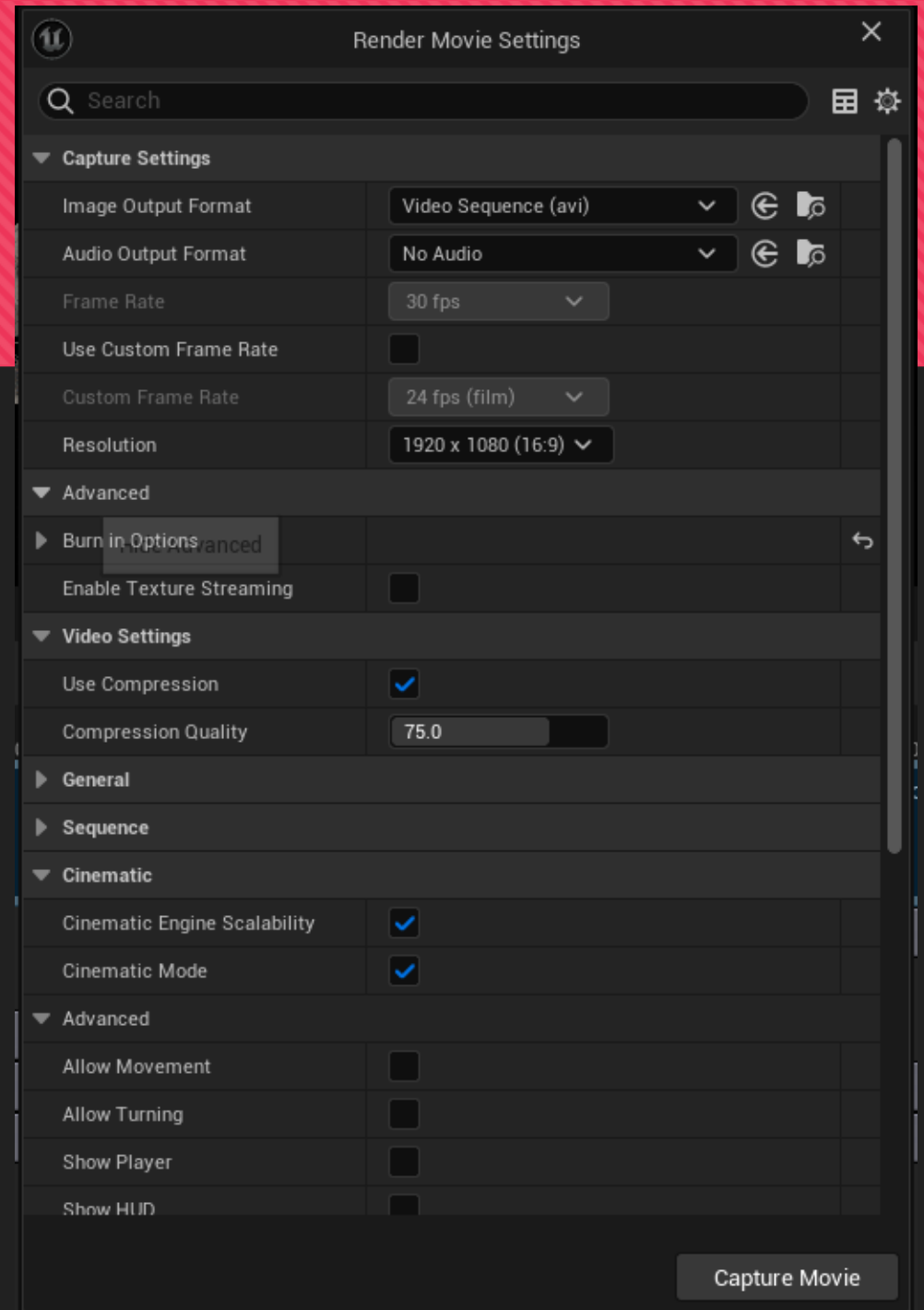
UE Docs: Old, Simple way:

<https://dev.epicgames.com/documentation/en-us/unreal-engine/old-render-movie-in-unreal-engine>



Render Movie

This is useful for quick video tests, but only exports low quality AVI sequences or individual frames



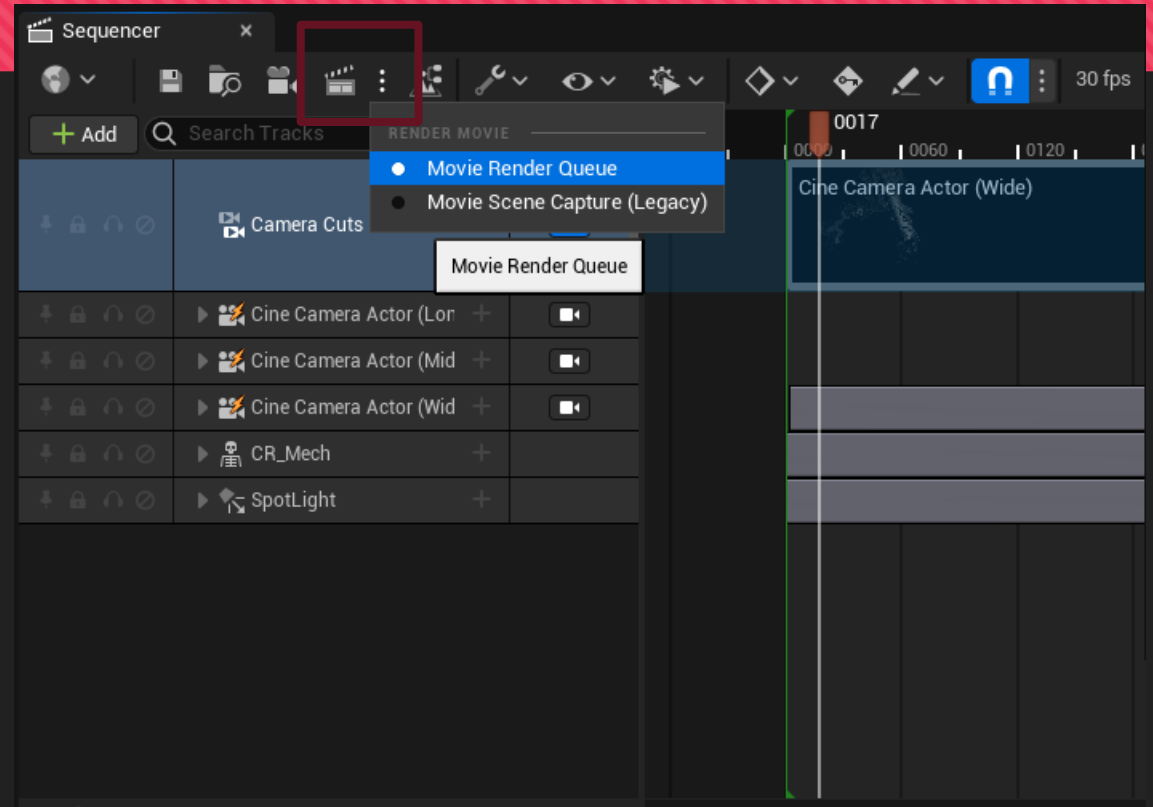
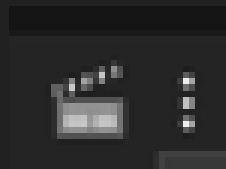
Movie Render Queue (newer Plugin)

For more power and control, we will explore the newer replacement – the Movie Render Queue.

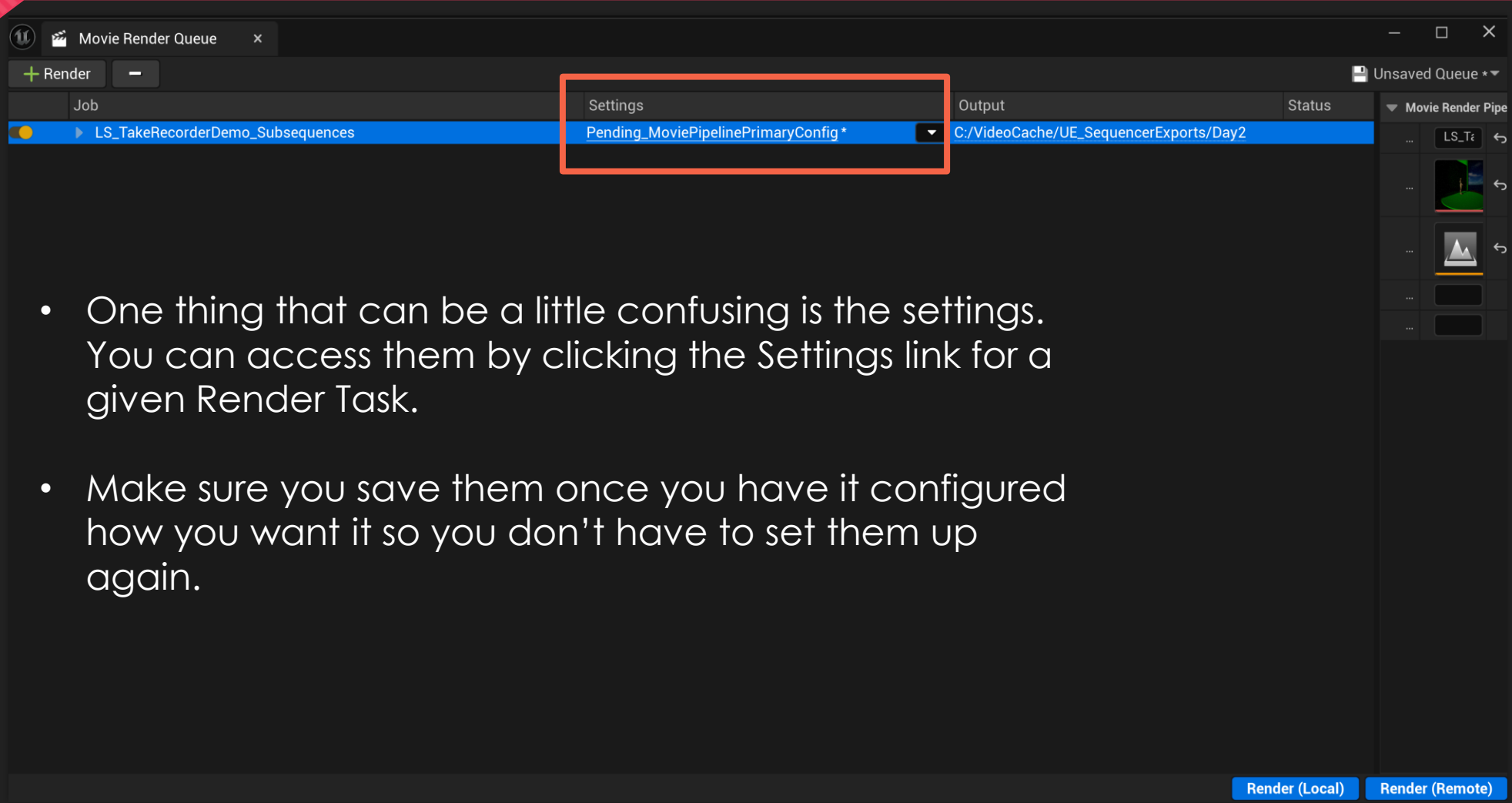
This brings many new features including better options and settings, control for exporting individual layers or render passes (e.g. Unlit, Lit, Detail) and a much higher quality output.

The Three Dots will open the selection menu for Movie Render Queue (new) or Movie Scene Capture (Legacy).

The Slate button will open the Render queue/MSD Window.



Render Queue Settings



Movie Render Queue

+ Render -

Job	Settings	Output	Status
LS_TakeRecorderDemo_Subsequences	Pending_MoviePipelinePrimaryConfig *	C:/VideoCache/UE_SequencerExports/Day2	

Unsaved Queue

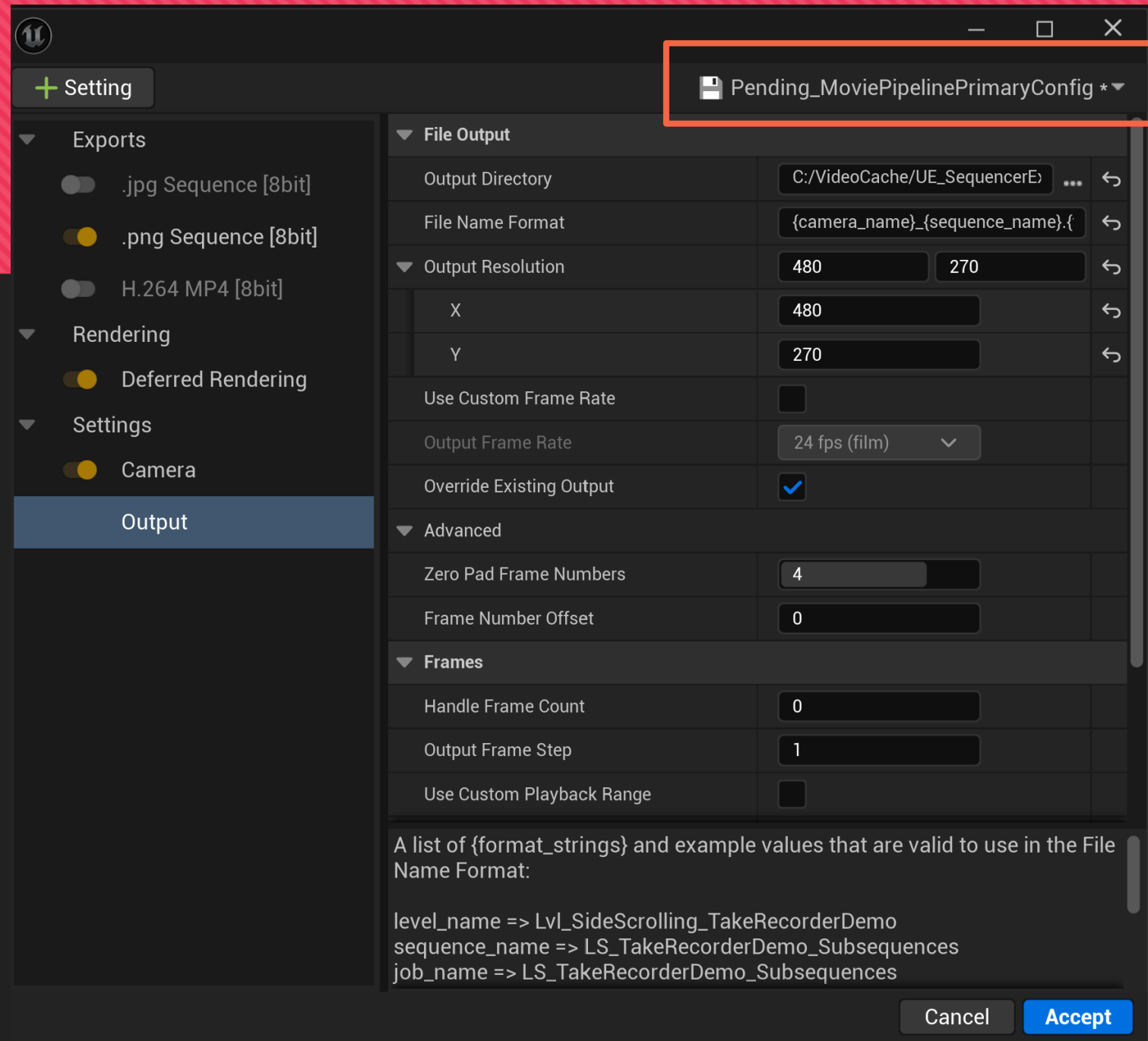
Movie Render Pipe

- One thing that can be a little confusing is the settings. You can access them by clicking the Settings link for a given Render Task.
- Make sure you save them once you have it configured how you want it so you don't have to set them up again.

Render (Local) Render (Remote)

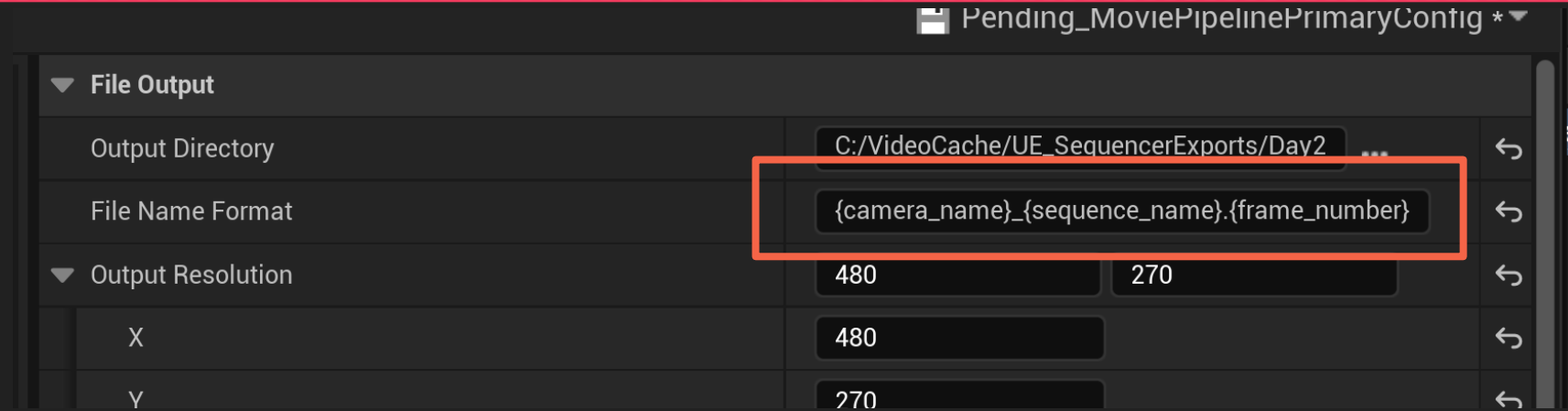
Settings - Saving

- Once you open the Setting panel, you can Save or Load the settings as a file in the top right corner.
- You can load previously saved settings here too.



Dynamic File Naming

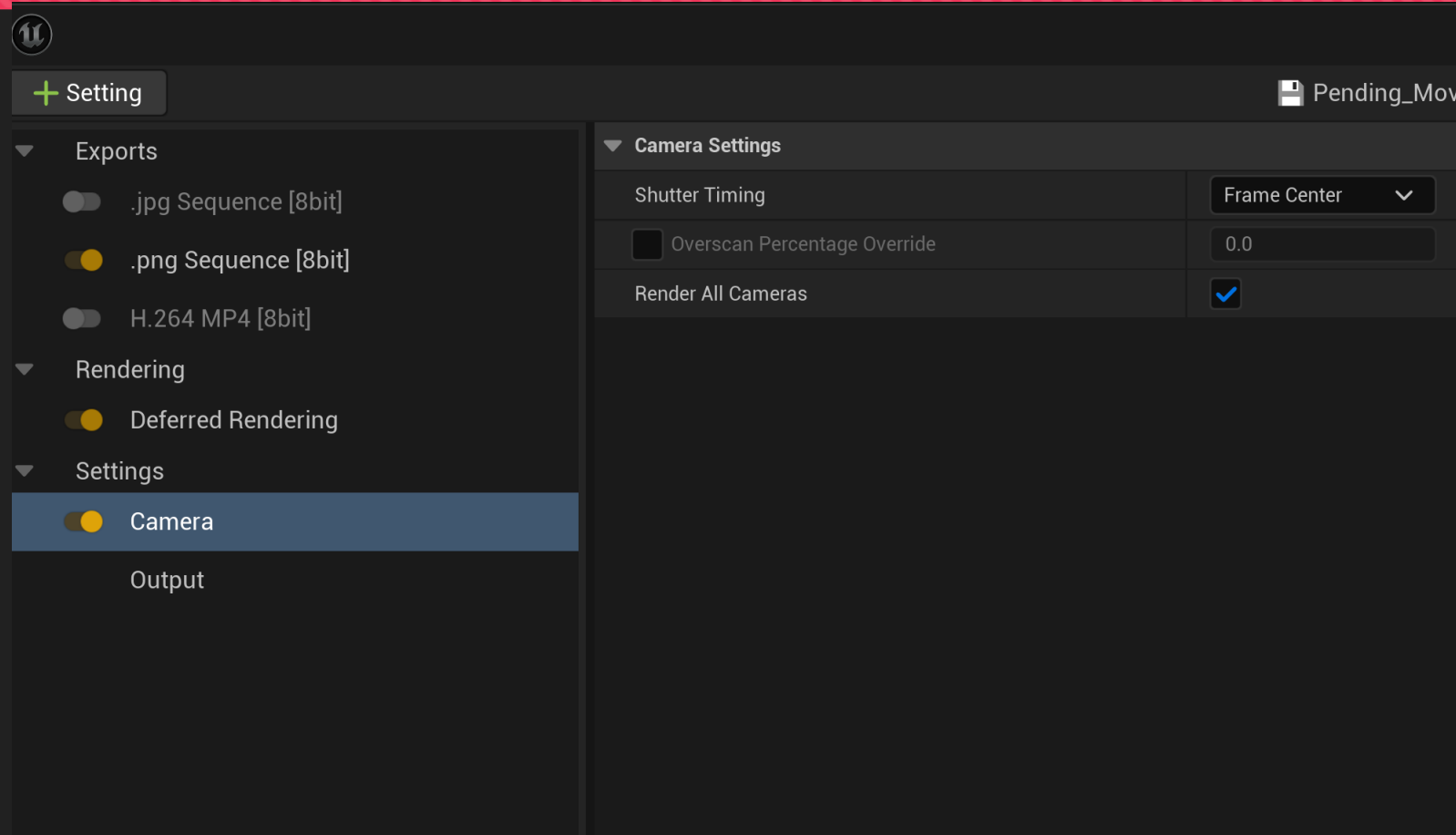
- You can use “Variables” to export useful information into the file name.
- In this example, I’m exporting the Camera Name , sequence name, and frame number.
- use the {camera_name} {sequence_name} and {frame_number} for this.
- There are many other options but this is a good start.



	CCA1_LS_TakeRecorderDemo_Subsequences.0168.png	2026-03-19 7:36 PM	PNG File
	CCA1_LS_TakeRecorderDemo_Subsequences.0169.png	2026-03-19 7:36 PM	PNG File
	CCA1_LS_TakeRecorderDemo_Subsequences.0170.png	2026-03-19 7:36 PM	PNG File
	CCA2_LS_TakeRecorderDemo_Subsequences.0000.png	2026-03-19 7:36 PM	PNG File
	CCA2_LS_TakeRecorderDemo_Subsequences.0001.png	2026-03-19 7:36 PM	PNG File
	CCA2_LS_TakeRecorderDemo_Subsequences.0002.png	2026-03-19 7:36 PM	PNG File
	CCA2_LS_TakeRecorderDemo_Subsequences.0003.png	2026-03-19 7:36 PM	PNG File

Render All Cameras

- To render each camera individually, add a Camera Settings option (in the +Setting menu), then check Render All Cameras.
- You'll need to name each camera differently with the variable tags for this to work (see previous slide) otherwise it will overwrite your previous camera passes.
- Note: Camera won't be in the Setting Menu if you have already added one.



Movie Render Queue

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Movie Render Queue

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+ Render

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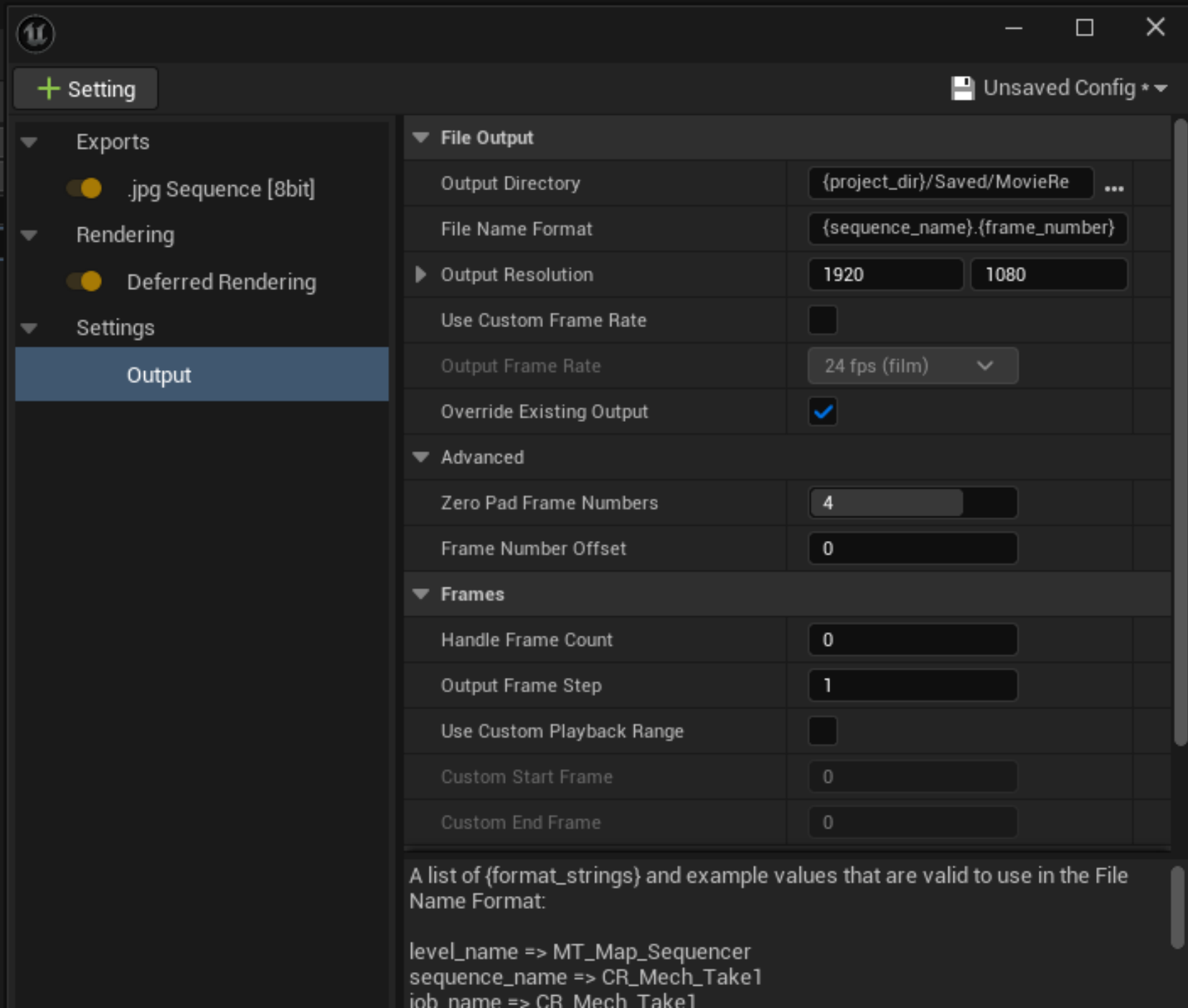
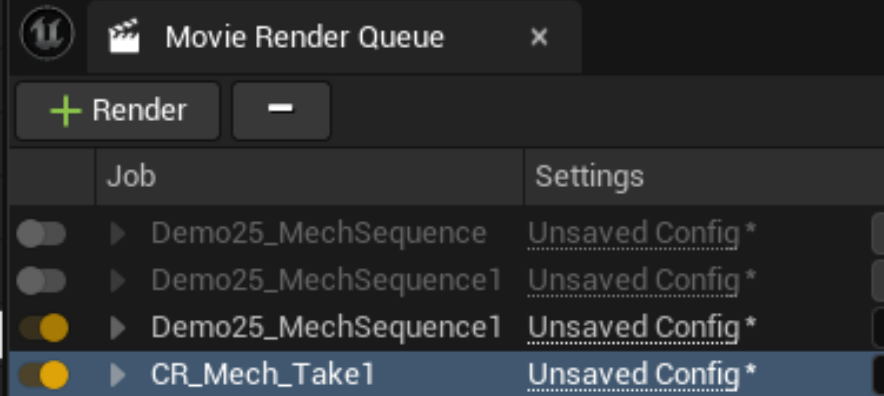
Job	Settings	Output	Status
▼ Demo25_MechSequence	Unsaved Config *	▼ C:/VideoCache/Unreal/MegaFPS	
● no shot(1) Cine Camera A Edit		▼	Ready
● no shot(2) Cine Camera A Edit		▼	Ready
● no shot(3) Cine Camera A Edit		▼	Ready
▶ Demo25_MechSequence1	Unsaved Config *	▼ {project_dir}/Saved/MovieRenders/	
● ▶ Demo25_MechSequence1	Unsaved Config *	▼ C:/VideoCache/Unreal/CameraComparison1	

Unsaved Queue * ▼

▼ Movie Render Pipeline

Job Name	Demo25_MechSequence	↩
Sequence	 Demo25_MechSequence ▼	↩
Map	 MT_Map_Sequencer ▼	↩
Author	Dylan	↩
Comment	First Try	↩

Render (Local) Render (Remote)



- Here, I am setting up the various settings available for rendering.
- I can choose multiple Export file types, which render passes to include, and the output settings.
- Note: The Render Queue can only export sequences of frames currently, and needs to be turned into a video in another application (I will be using DaVinci Resolve.)

Setting up FFMPEG so I can render out MP4s

The MP4 export won't work without this.

Mine didn't work even with it, but maybe you can figure it out.

Dylan

Plugins - Movie Pipeline CLI Encoder

Movie Pipeline Command Line Encoder Settings

Export...

Import...

These settings are saved in DefaultEngine.ini, which is currently writable.

Command Line Encoder

Executable Path	"C:\Program Files\FFMPEG\ffmpeg-2026-03-18-git-106616f13d-essentials_build\bin\ffmpeg.exe"
Codec Help Text	Run 'MovieRenderPipeline.DumpCLIEncoderCodecs' in Console to see available codecs. 
Video Codec	libx264
Audio Codec	aac
Output File Extension	mp4

Command Line Arguments

While there is a Video output option (Mp4), it may require some setup to get it working correctly, including linking the command line FFMPEG library.

Feel free to do this if you'd like, as it will save you the trouble of importing frames into DaVinci, but I found it problematic, and the renders often failed anyways.

The other benefit of rendering PNG frames is that if the rendering crashes or stalls, you won't lose the whole sequence, and you can just re-render specific frames you might need.

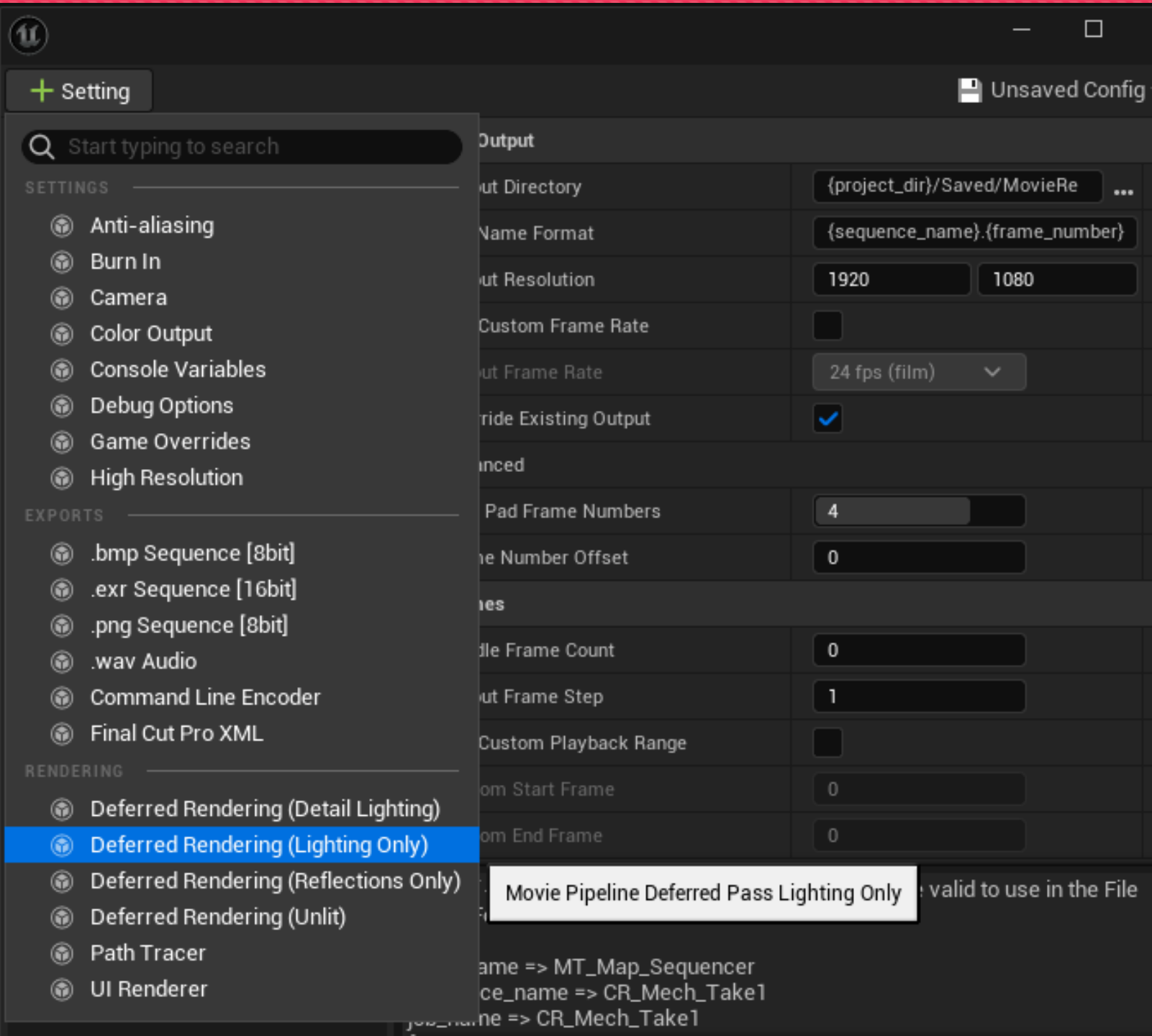
Output Log

- You can check the Message Log for warnings or errors as well.
- This system can be a bit finicky, so check out the Troubleshooting and Tips at the end of these slides before getting started.



Lighting Passes

You can choose different lighting passes, which is extremely useful for troubleshooting, as well as VFX work.



Master

Start TC	Date Created	Clip Name
	Sat Jan 25 2025 22:49:12	MegaFPS
01:00:00:...	Sat Jan 25 2025 22:55:04	Timeline 1
	Mon Jan 27 2025 11:02:19	MegaFPS Mech2
01:00:00:...	Mon Jan 27 2025 11:23:24	Timeline 2 - Mech Camera Comparison

Create New Timeline... Ctrl+N
Create New Timeline Using Selected Bin...

New Bin

Import Media... Ctrl+I

Remove All Clips in Bin...
Relink Clips for Bin...

DaVinci Resolve

We will keep things very simple
for now.

Create a new Bin (think of it like
a folder)

DaVinci Resolve

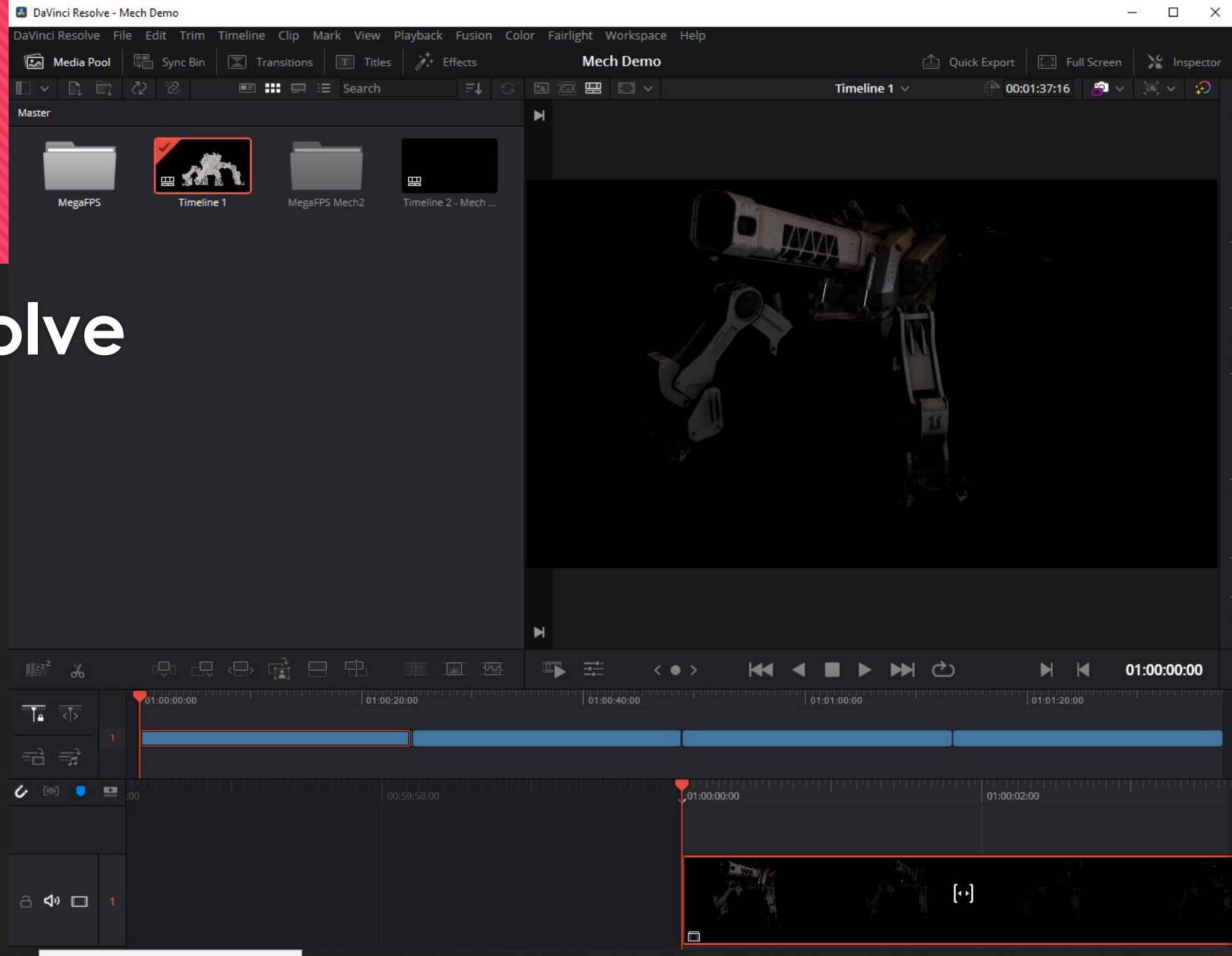
- We will use DaVinci to import the Frames that Unreal exported, and turn it into a Video clip.
- DaVinci Resolve is great as it is a professional level video editing and color grading suite. It offers a robust Free version as well. It is both accessible and extremely powerful, on par with any other edit suite on the market.
- We will just be using some basic tools in it though for now, but you could fully edit your movie in DaVinci Resolve if you wanted it.

DaVinci Resolve

Create a Timeline

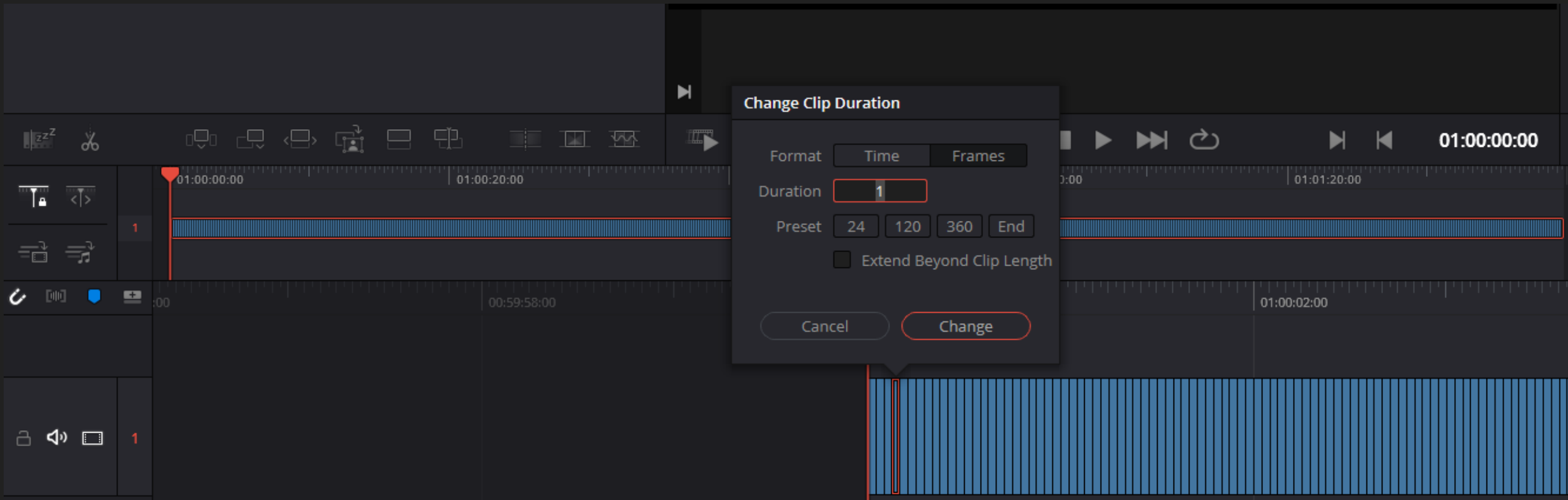
This will be the actual sequence to edit/turn into video.

Drag your exported frames into the Bin, then select all of them, and drag them from the Bin into the Timeline.



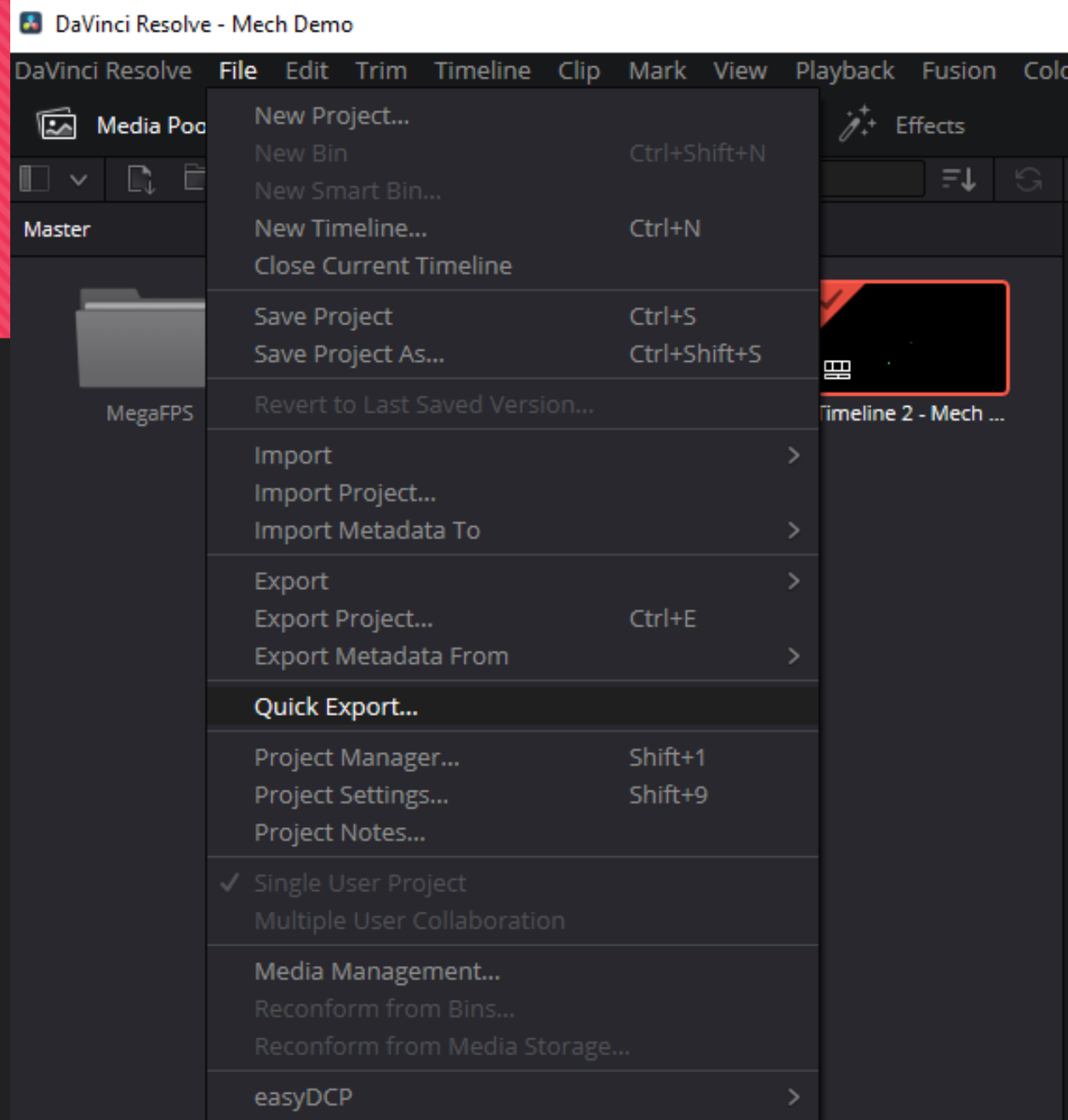
Change Clip Length

Within the Timeline, select all clips, R-Click and set the Duration of the Clip to 1 Frame each. This will play the sequence we exported at a normal speed now.



DaVinci Export

Select File > Quick Export with
your Timeline open.



Quick Export

H.264

H.264 Master

H.264

HyperDeck

H.265

H.265 Master



YouTube



Vimeo



Twitter



Dropbox

Filename

Timeline 2 - ...ra Comparison

Resolution

1920 x 1080

Video Codec

H.264

Frame Rate

24 fps

Audio

Stereo

Audio Codec

AAC

Duration

1m 40s

Choose your export settings.
H.264 will be large and
uncompressed, but high quality.
You might want to export
something smaller to test at first.

Probably some amount of
compression will be necessary
to get your files to a reasonable
size.

Cancel

Export

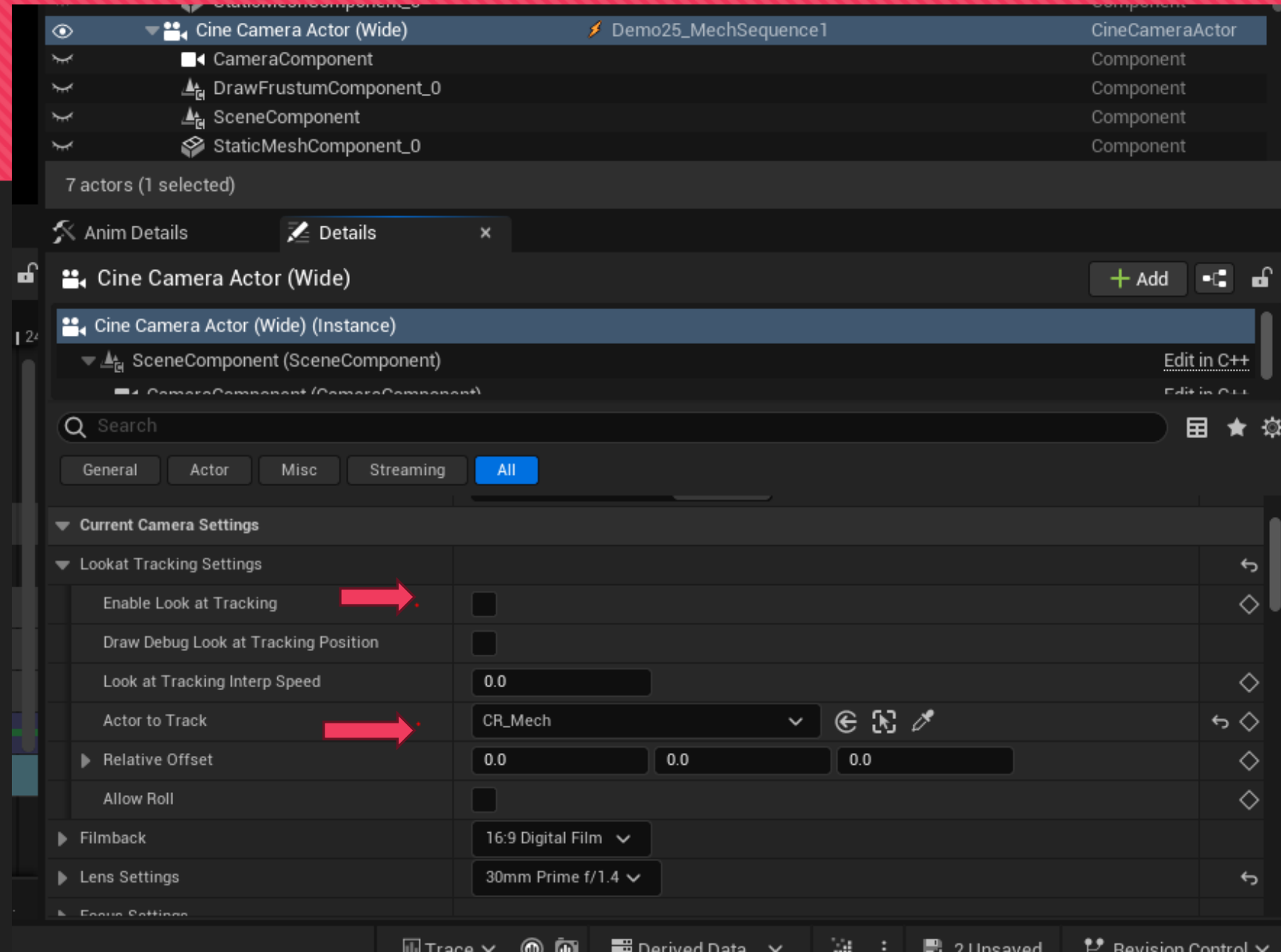
Misc Tips & Troubleshooting

Cine Camera Tracking

To have the Cine Camera follow a target as it moves, set up the Lookat Tracking Settings.

Tip: You might want to use the Relative Offset to help frame a more interesting shot (rather than staring straight at your subject).

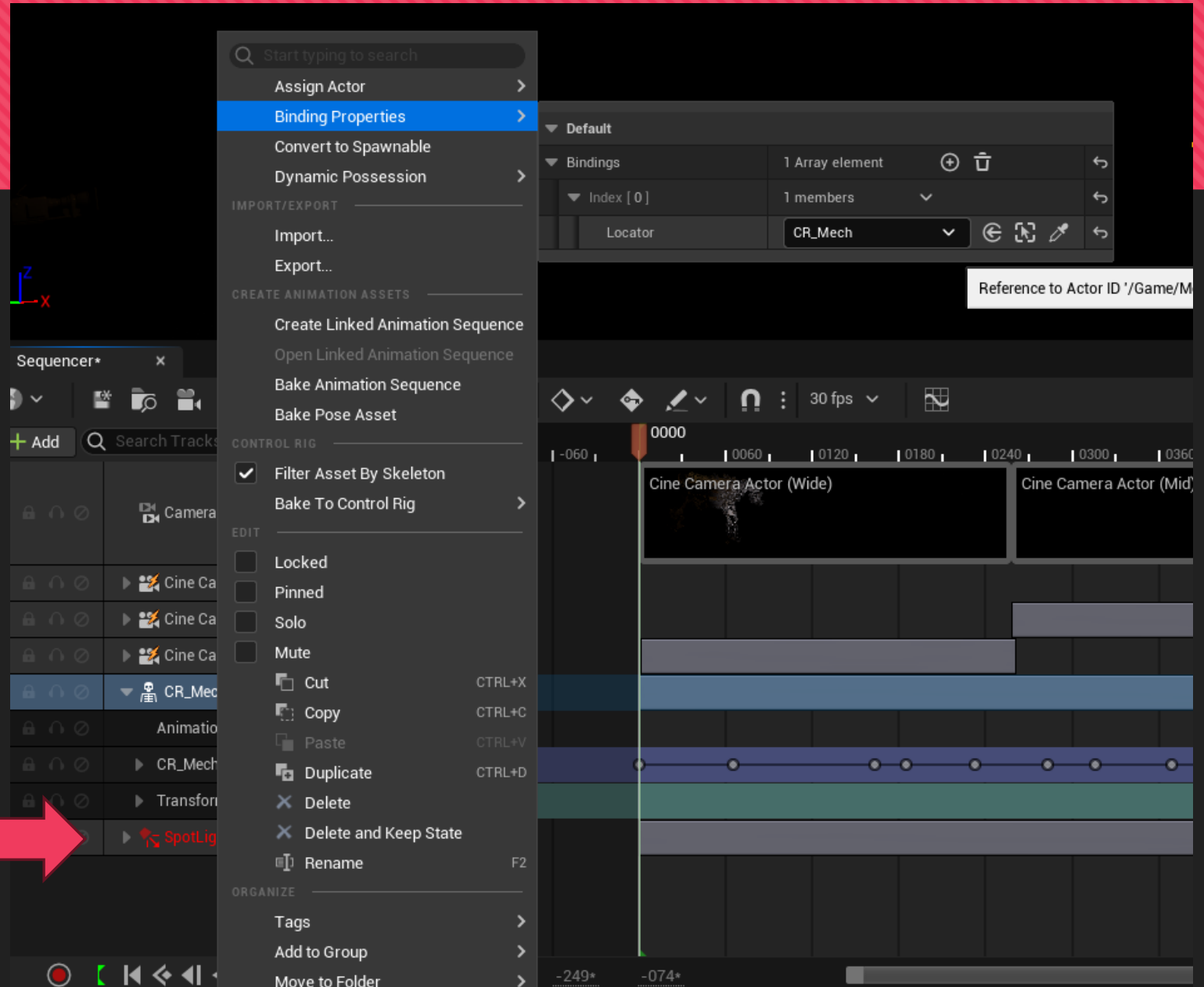
You can also create another “Focus” object and use that as a target instead. Just set its visibility to Off. This can be helpful if you are trying to look at something large or awkwardly shaped.



Sequencer Rebinding

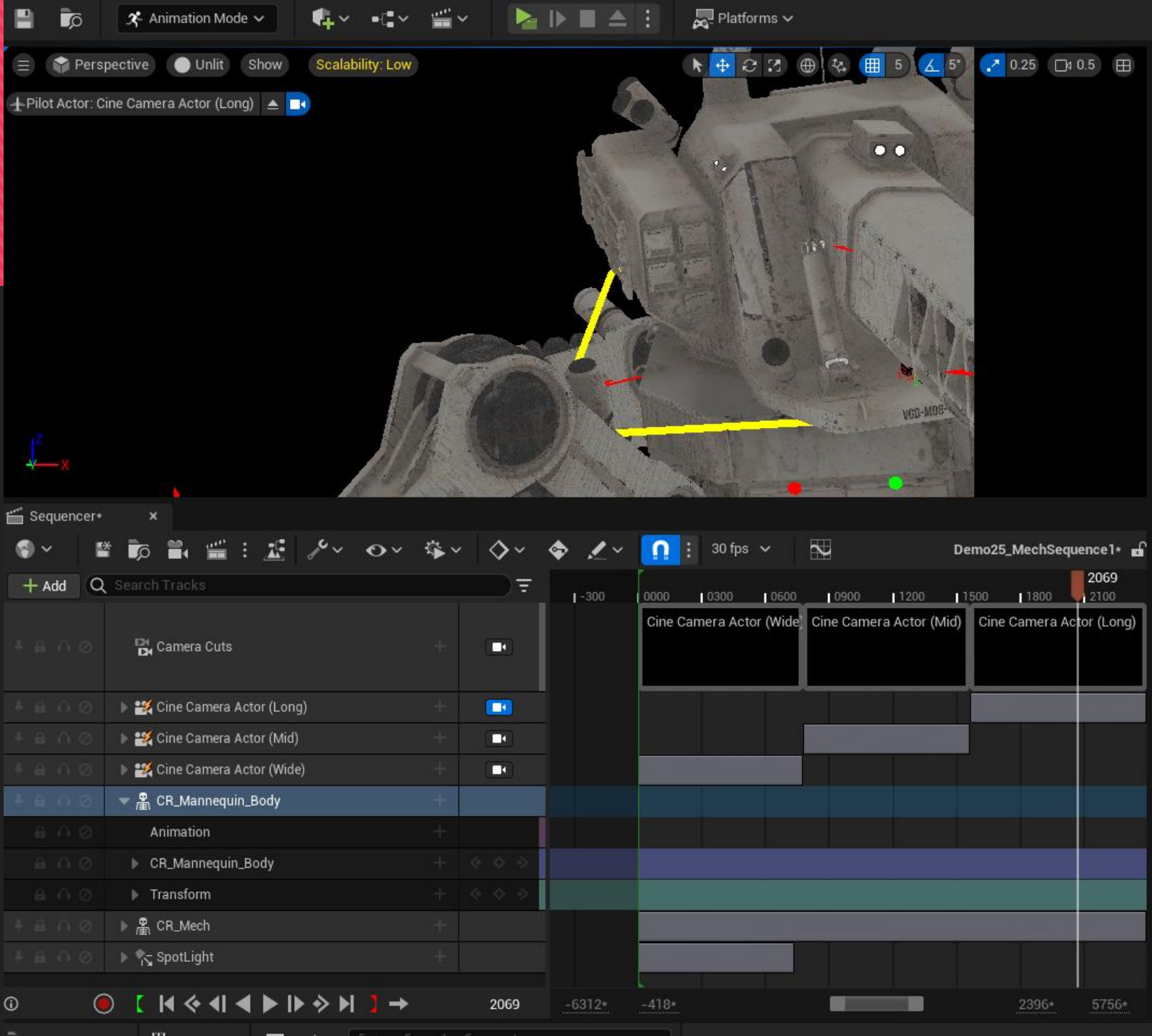
Tip: If your Model in the Sequencer gets disconnected (turns Red) you can reconnect it to the asset using the Binding Properties option in the R-Click Menu

You can see the Spotlight is currently missing its asset reference. This can occur if you delete objects from the scene that were being referenced by the Sequencer.



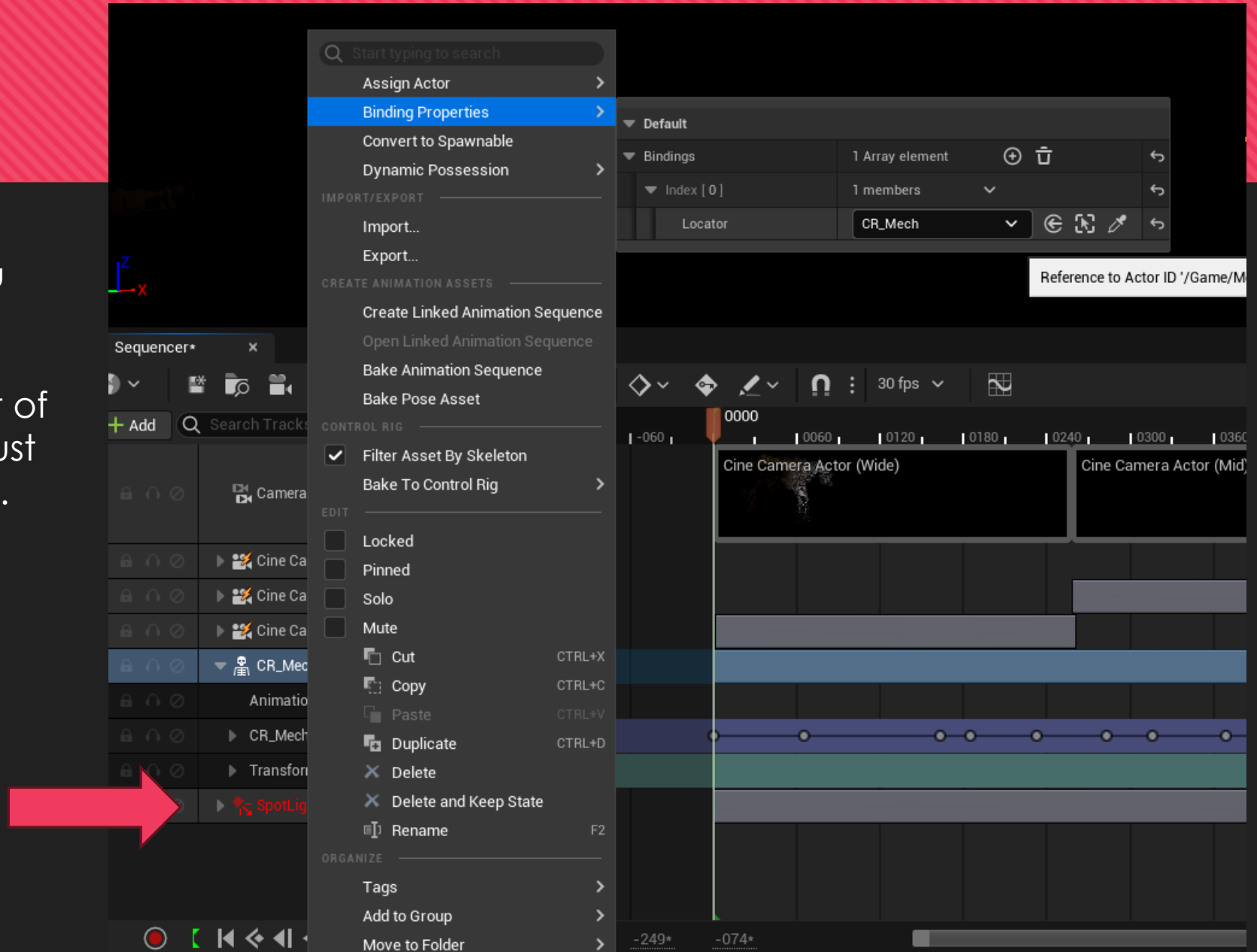
Sequencer Rebinding.

Notice the CR Mannequin Body is now attached to the source asset again.



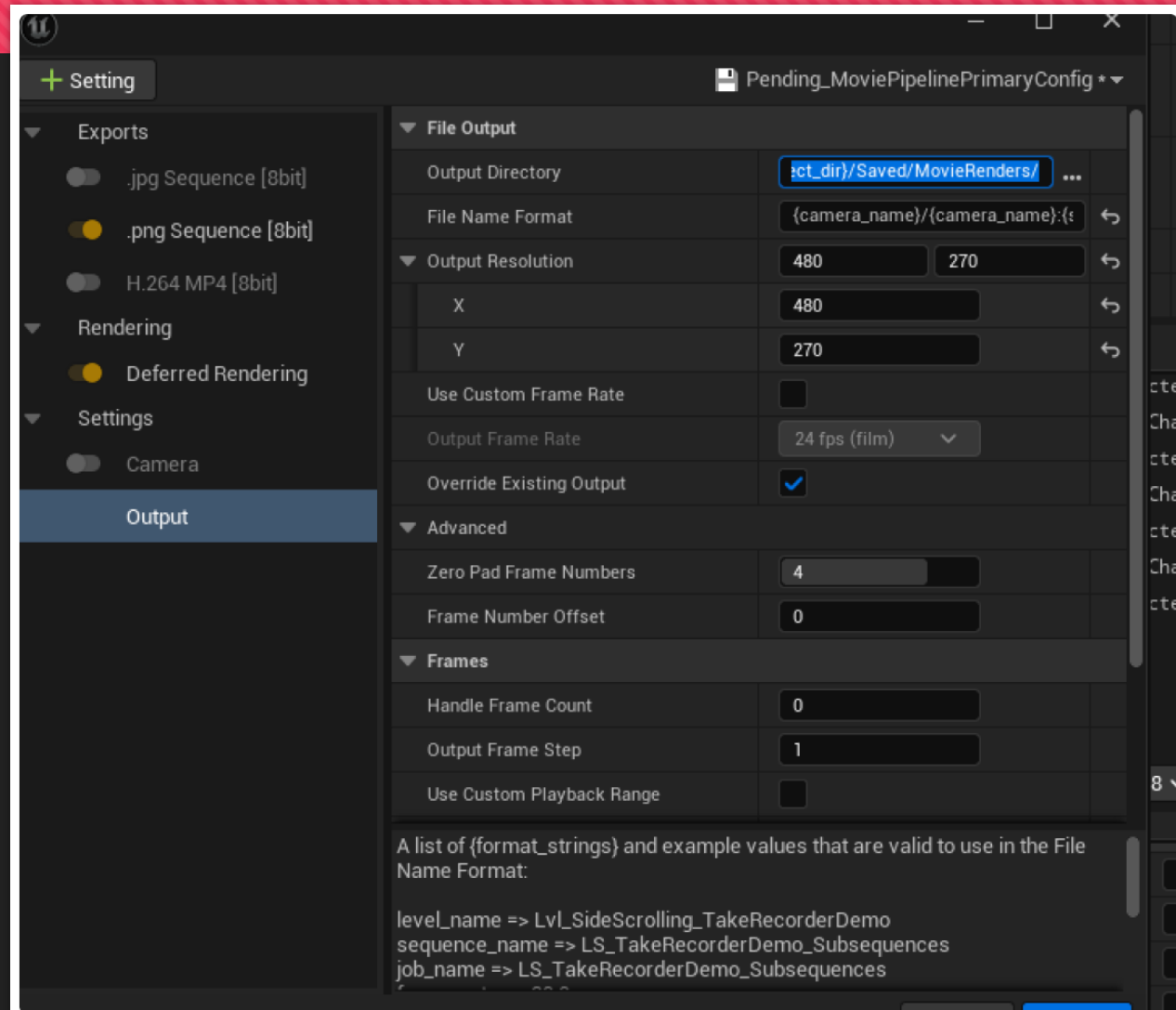
Sequencer Rebinding

Tip: This also might mean you opened a Level Sequence outside of the level it was created in, so if you see a lot of objects turn red, you might just need to open the Level itself.



Watch Your Spaces

- I ran into a lot of weird issues trying to export for the first time. The renderer will break on many simple issues with the file path or naming:
- Any space in the output name (this was tricky because by default it was {project_dir} which itself had a space in it).
- I created a simpler path to render to and that fixed it.
- Special characters in the name ":" broke the name. Use _
- Periods "." can work but it can sometimes confuse the file type, so underscore is recommended.



Watch Your Spaces

- I ran into a lot of weird issues trying to export for the first time. The renderer will break on many simple issues with the file path or naming:
- Admin Access: I required Admin access to the folder I was rendering to, so you might have to run UE in Admin mode.
- If your camera names are too long, or the same, it can also cause issues, so rename them to something short and distinct.

